

$$\frac{\partial u_i}{\partial x_j} \cdot v_j = v_j \cdot \frac{\partial}{\partial x_j} (u_i) = \left(v_j \cdot \frac{\partial}{\partial x_j} \right) u_i$$

$$z_i = \underbrace{\left(v_1 \cdot \frac{\partial (\cdot)}{\partial x_1} + v_2 \cdot \frac{\partial (\cdot)}{\partial x_2} + v_3 \cdot \frac{\partial (\cdot)}{\partial x_3} \right)}_{\Delta} u_i = \Delta u_i$$

$$z_1 = v_1 \frac{\partial u_1}{\partial x_1} + v_2 \frac{\partial u_1}{\partial x_2} + v_3 \frac{\partial u_1}{\partial x_3}$$

$$z_2 = v_1 \frac{\partial u_2}{\partial x_1} + v_2 \frac{\partial u_2}{\partial x_2} + v_3 \frac{\partial u_2}{\partial x_3}$$

$$z_3 = \dots$$

$$\Delta = \underline{v} \cdot \underline{\nabla} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \frac{\partial}{\partial x_3} \end{bmatrix} = v_1 \cdot \frac{\partial (\cdot)}{\partial x_1} + v_2 \frac{\partial (\cdot)}{\partial x_2} + v_3 \cdot \frac{\partial (\cdot)}{\partial x_3}$$

— 0 —

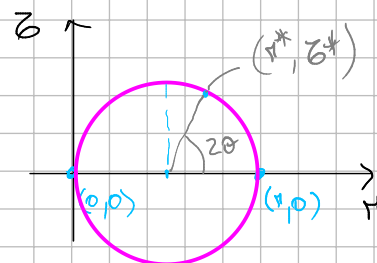
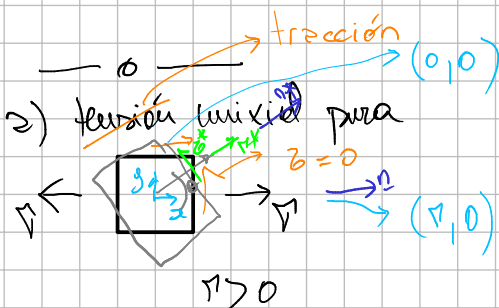
$= \varepsilon_{jki} A_{1j} A_{2k} A_{3i} \checkmark$

2) $r^2 = x_i x_i$, $f(r)$ es una función arbitraria de r

a) $\nabla(f(r)) = \frac{\partial}{\partial x_j} (f(r)) \mathbf{i} = f'(r) \cdot \frac{\partial r}{\partial x_j} = f'(r) \frac{x_j}{r} = f'(r) \frac{\mathbf{x}}{r} \checkmark$

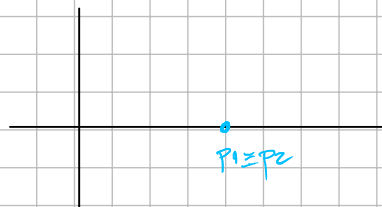
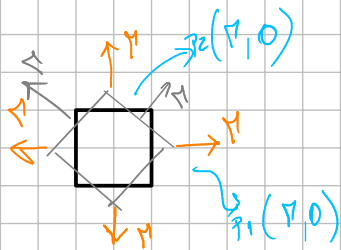
b) $\nabla^2(f(r)) = \nabla \cdot (\nabla(f(r)))$
 $= \nabla \cdot (f'(r) \frac{\mathbf{x}}{r})$
 $= \frac{\partial}{\partial x_i} \left(f'(r) \frac{x_i}{r} \right)$
 $= f''(r) \cdot \frac{\partial r}{\partial x_i} \cdot \frac{x_i}{r} + f'(r) \frac{\partial}{\partial x_i} \left(\frac{x_i}{r} \right)$
 $= f''(r) \cdot \frac{x_i}{r} \cdot \frac{x_i}{r} + f'(r) \cdot \left(\frac{1}{r} \right)$

$$\begin{aligned} & \frac{\partial}{\partial x_i} \left(\frac{x_i}{r} \right) = \frac{\partial (x_i)}{\partial x_i} \cdot \frac{1}{r} + x_i \cdot \frac{\partial}{\partial x_i} \left(\frac{1}{r} \right) \\ &= \frac{3}{r} + x_i \cdot (-1) \cdot \frac{1}{r^2} \cdot \frac{\partial r}{\partial x_i} \\ &= \frac{3}{r} - \frac{x_i \cdot x_i}{r^3} = \frac{2}{r} \end{aligned}$$

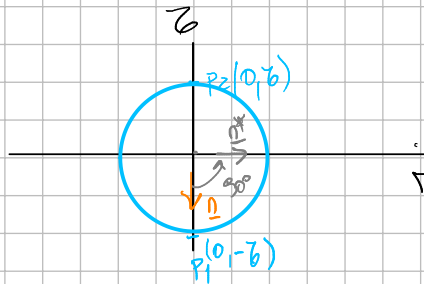
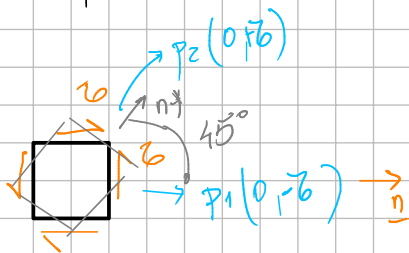


$$\tau^* = \sigma \cdot \mathbf{n} = \sigma \cdot \mathbf{n}^*$$

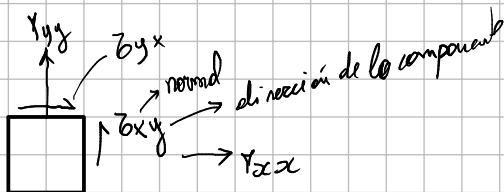
c) broca de bixid pura



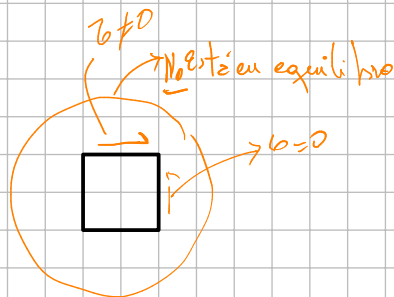
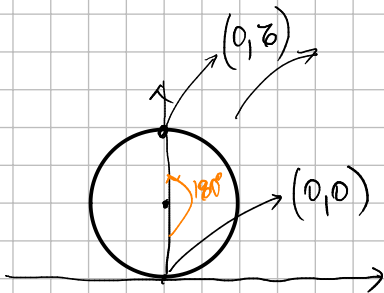
d) lote puro



$$\tau_{ij} = \tau_{ji}$$



Bal de cont. de mov. angular
(equilibrio de momentos)
 $\tau_{xy} = \tau_{yx}$



a. $\nabla \cdot (r^n \underline{r}) = (n+3)r^n$

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

$$r = \|\underline{r}\| \Rightarrow r^2 = \underline{r} \cdot \underline{r}$$

$$\nabla \cdot (r^n \underline{r}) = \frac{\partial}{\partial x_i} (r^n r_i) = n r^{n-1} \cdot \frac{\partial r}{\partial x_i} \cdot r_i + r^n \cdot \frac{\partial (r_i)}{\partial x_i}$$

$$\underline{r} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} \frac{\partial r}{\partial x_i} &= \frac{\partial \sqrt{\underline{r} \cdot \underline{r}}}{\partial x_i} = \frac{\partial (\sqrt{x_j \cdot x_j})}{\partial x_i} = \frac{1}{2} \frac{1}{\sqrt{x_j \cdot x_j}} \cdot \frac{\partial (x_j \cdot x_j)}{\partial x_i} \\ &= \frac{1}{2} \frac{1}{\sqrt{x_j \cdot x_j}} \cdot \left(\frac{\partial x_j}{\partial x_i} \cdot x_j + x_j \cdot \frac{\partial x_j}{\partial x_i} \right) \\ &= \frac{1}{2} \frac{1}{r} \cdot (x_i + x_i) = \frac{1}{2} \frac{1}{r} \cdot 2x_i = \frac{x_i}{r} \end{aligned}$$

- $r^2 = r_j \cdot r_j = x_j \cdot x_j$

$$\frac{\partial(r^2)}{\partial x_i} = 2r \cdot \frac{\partial r}{\partial x_i}$$

$$\frac{\partial(x_j x_j)}{\partial x_i} = 2x_i$$

$$2r \cdot \frac{\partial r}{\partial x} = 2x_i \Rightarrow \frac{\partial r}{\partial x} = \frac{x_i}{r}$$

$\varphi \rightarrow$ composite color

$\varphi \Rightarrow$ composite residual

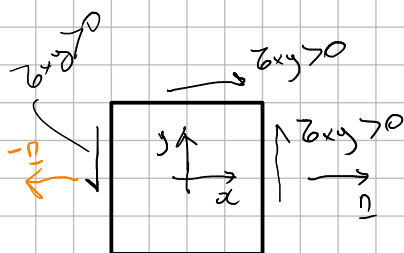
$$u_i = \frac{\partial \varphi(\underline{x})}{\partial x_i} =$$

$$\varphi(\underline{x}) = \ln(x_1^2) + x_2^3 \cdot \ln(x_3)$$

$$\frac{\partial \varphi(\underline{x})}{\partial x_1} = \frac{\partial}{\partial x_1} (\ln x_1^2) + \frac{\partial}{\partial x_1} (x_2^3 \cdot \ln(x_3))$$

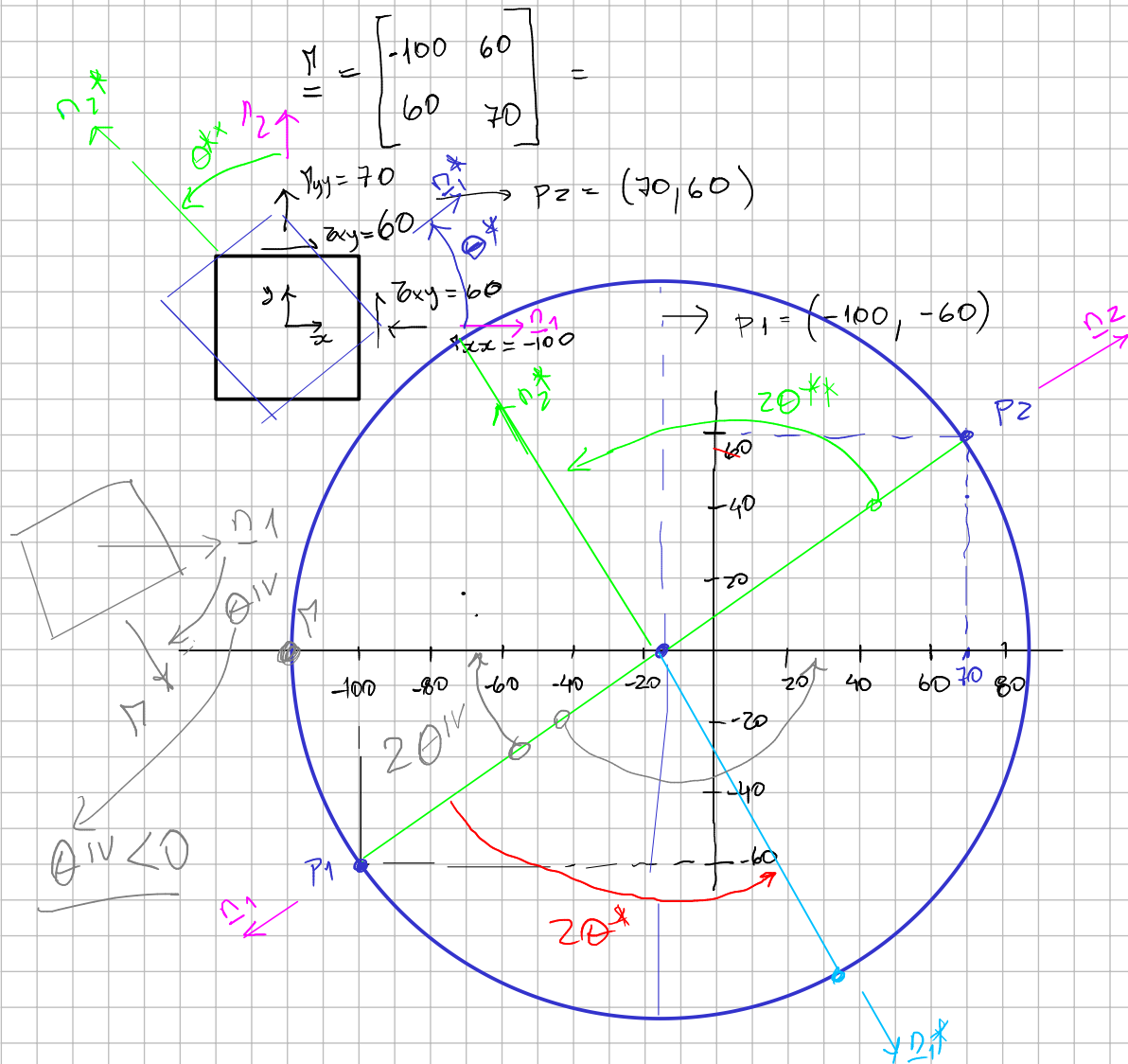
$$\frac{\partial \varphi(\underline{x})}{\partial x_2} = \frac{\partial}{\partial x_2} (\ln x_1^2) + \frac{\partial}{\partial x_2} (x_2^3 \cdot \ln(x_3))$$

$$\ln(x_3) \cdot 3x_2^2 \cdot \frac{\partial x_2^3}{\partial x_2}$$



$\bar{x}y > 0$

$$\pi = \begin{bmatrix} r_{xx}(\bar{x}y) \\ \bar{x}y \quad r_{yy} \end{bmatrix}$$



$$\underline{\underline{\gamma}} = \begin{bmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & \sigma \end{bmatrix} = \sigma \cdot \underline{\underline{I}} \rightarrow \text{isotropic}$$

$$\underline{\underline{\gamma}}' = \underline{\underline{P}} \underline{\underline{\gamma}} \underline{\underline{P}}^T = \underline{\underline{P}} \sigma \underline{\underline{I}} \underline{\underline{P}}^T = \sigma \cdot \underline{\underline{P}} \underline{\underline{I}} \underline{\underline{P}}^T = \sigma \underline{\underline{P}} \underline{\underline{P}}^T$$

